



COS20028 – BIG DATA ARCHITECTURE AND APPLICATION LAB WEEK 01

Dr Pei-Wei Tsai (Lecturer, Tutor, and Unit Convenor)
ptsai@swin.edu.au, EN508d





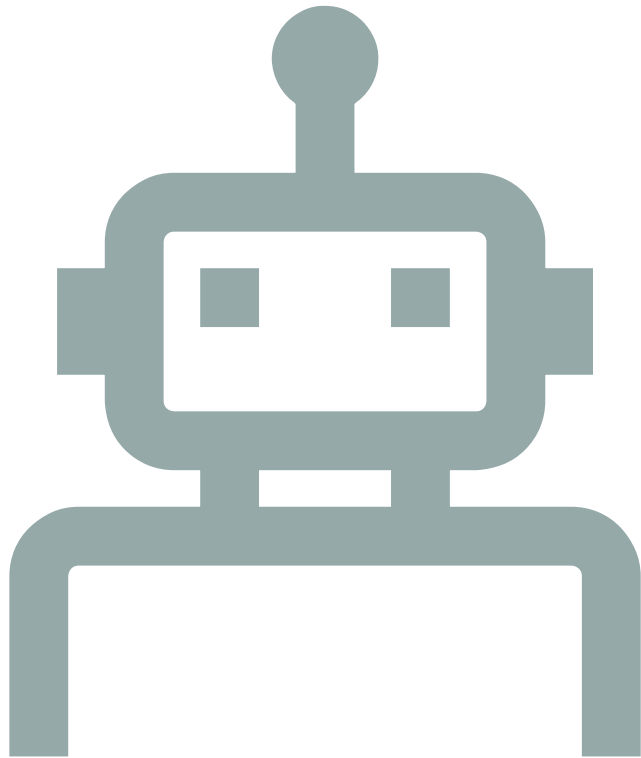
Week 1



VM Setup



Hadoop
Commands
Warm Up



What is a Virtual Machine?

- A virtual machine is a simulated environment to be hosted by your physical machine.
- It can have different Operating Systems (OS), applications, and hardware than your physical machine.
- It is usually used as a sandbox, an experimental environment, or an agent in practice.

VMware Workstation Player Setup



1. Install VMware Workstation Player or VMware Fusion.

Learn how to download following:

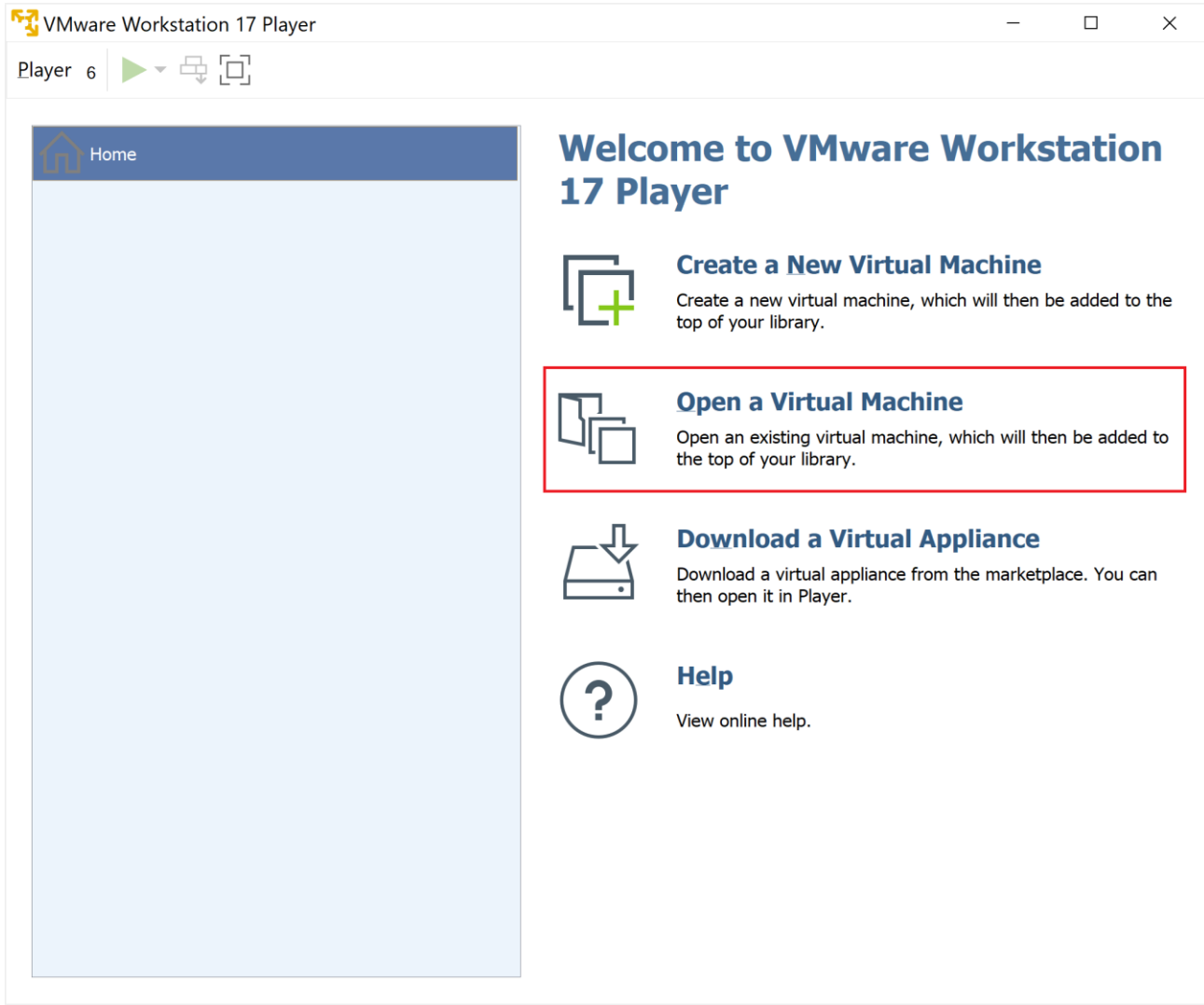
<https://knowledge.broadcom.com/external/article?articleNumber=368734>

or

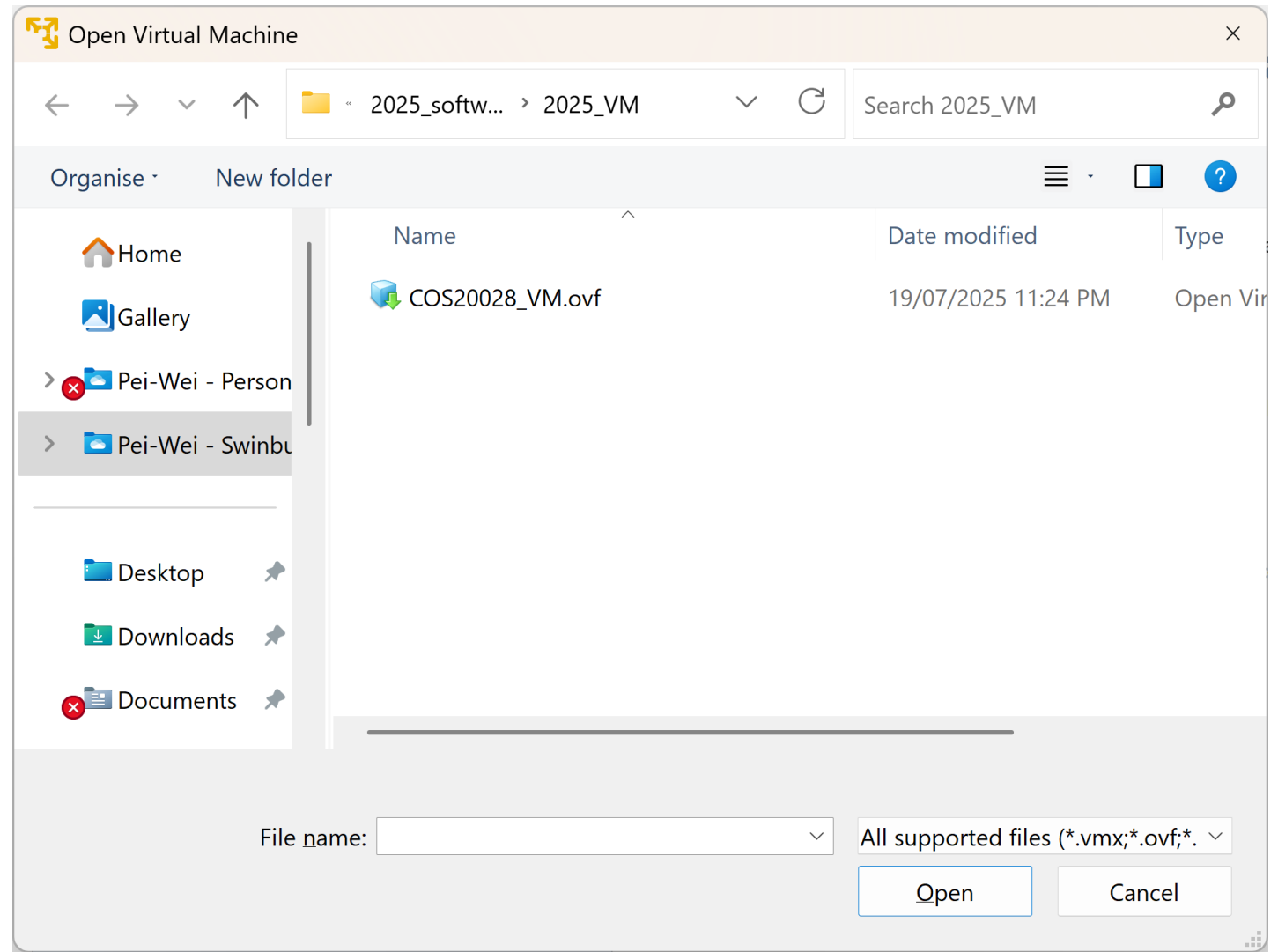
2. Download the VM Image in Canvas.

Module → COS20028 Tools → Cloudera VM

Import the VM Image

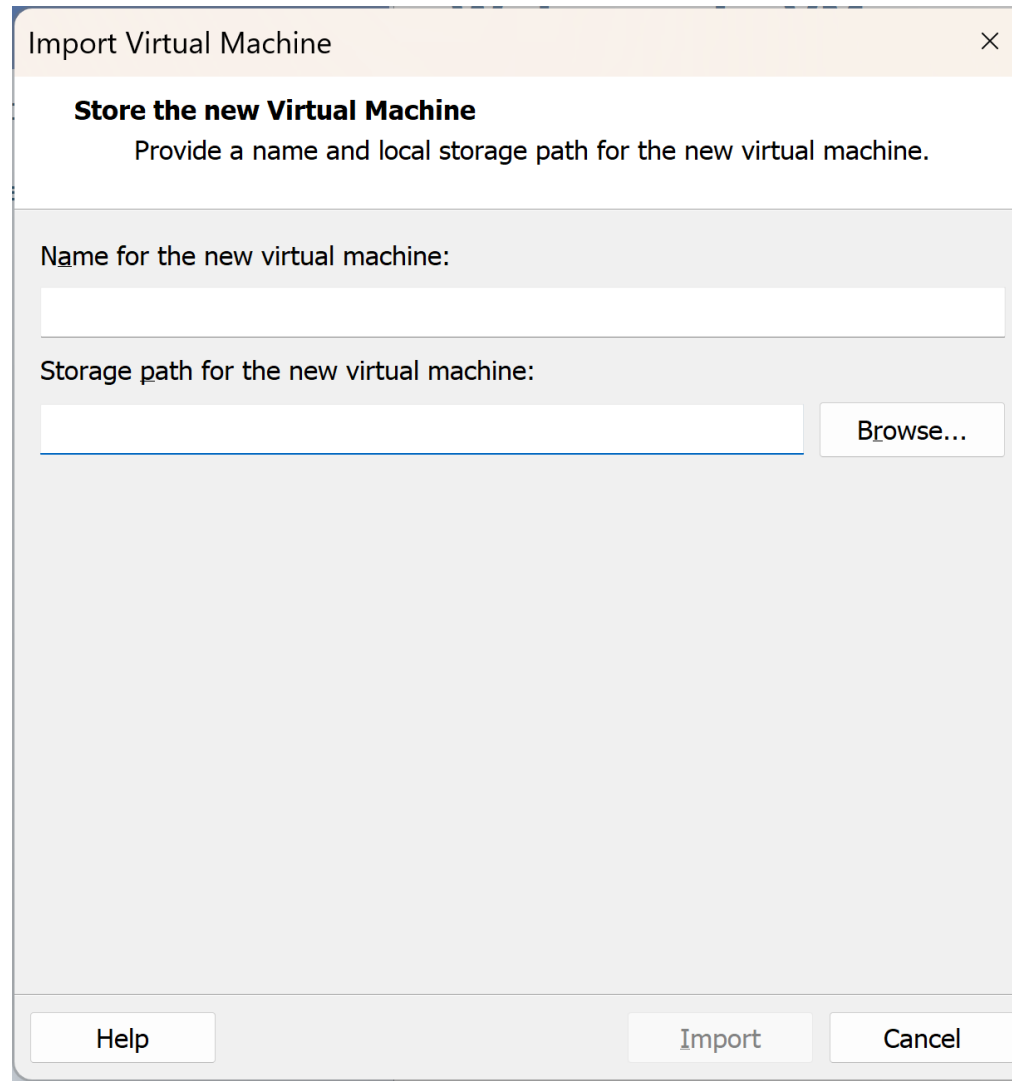


Select the Image



Name Your Virtual Machine

- Type in the "Name for the new virtual machine". The storage path will automatically be filled. Click "Import" to load the image.



The screenshot shows a dialog box titled "Import Virtual Machine" with a close button (X) in the top right corner. The main heading inside is "Store the new Virtual Machine", followed by the instruction "Provide a name and local storage path for the new virtual machine." Below this, there are two input fields. The first is labeled "Name for the new virtual machine:" and is empty. The second is labeled "Storage path for the new virtual machine:" and is also empty, with a "Browse..." button to its right. At the bottom of the dialog, there are three buttons: "Help", "Import", and "Cancel".

Let the Software Setup Your Machine

- Give it some time to load the selected image.

VMware Workstation 17 Player

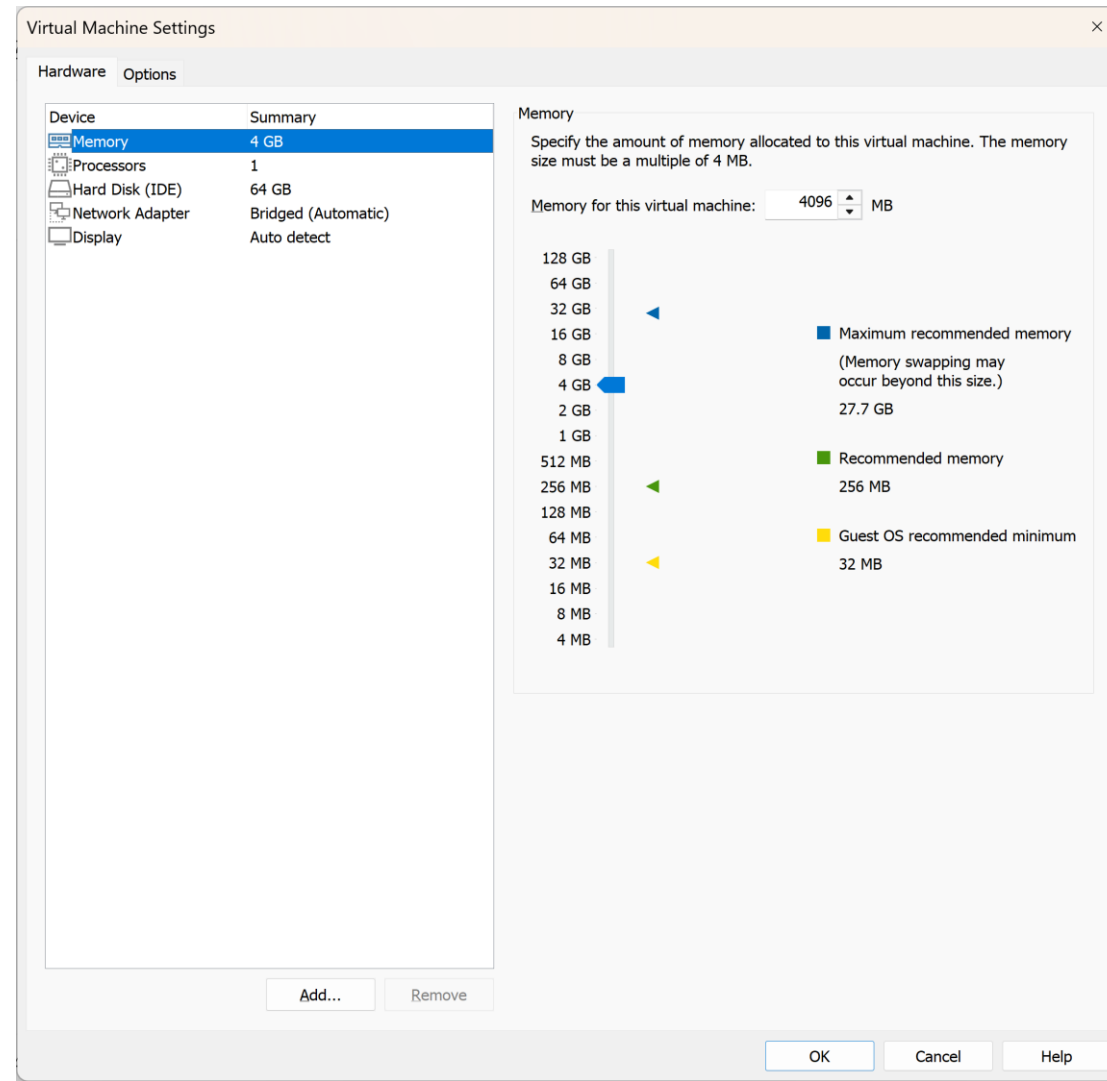
Importing COS20028_VM



Cancel

Setup Your VM, or Execute it.

- Double-click on the machine's name to start it. You can also use "Edit virtual machine settings" to change some resource-occupying details.



Oracle VM VirtualBox Setup



1. Install Oracle VM VirtualBox.

Download from

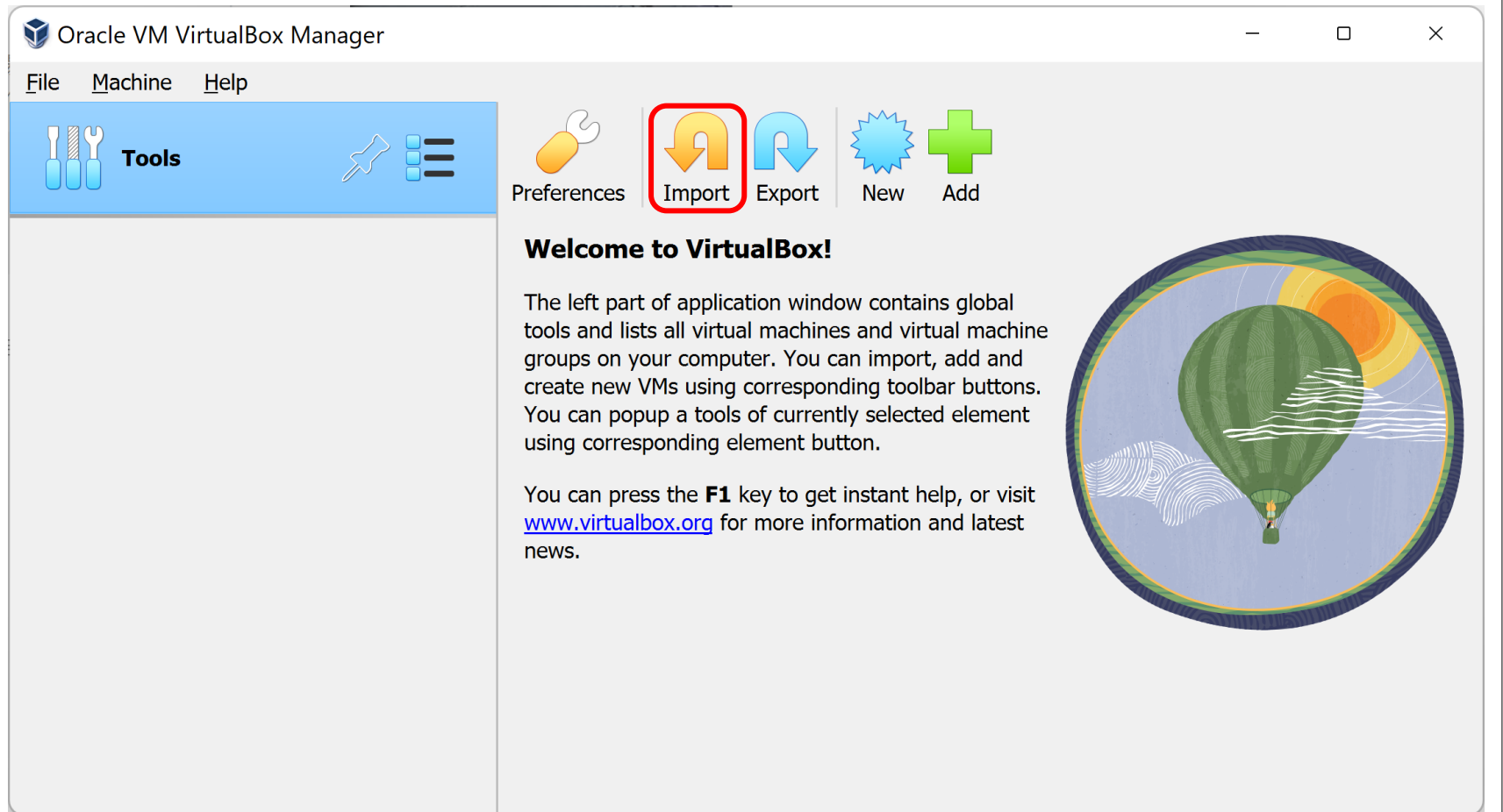
<https://www.virtualbox.org/wiki/Downloads>

or

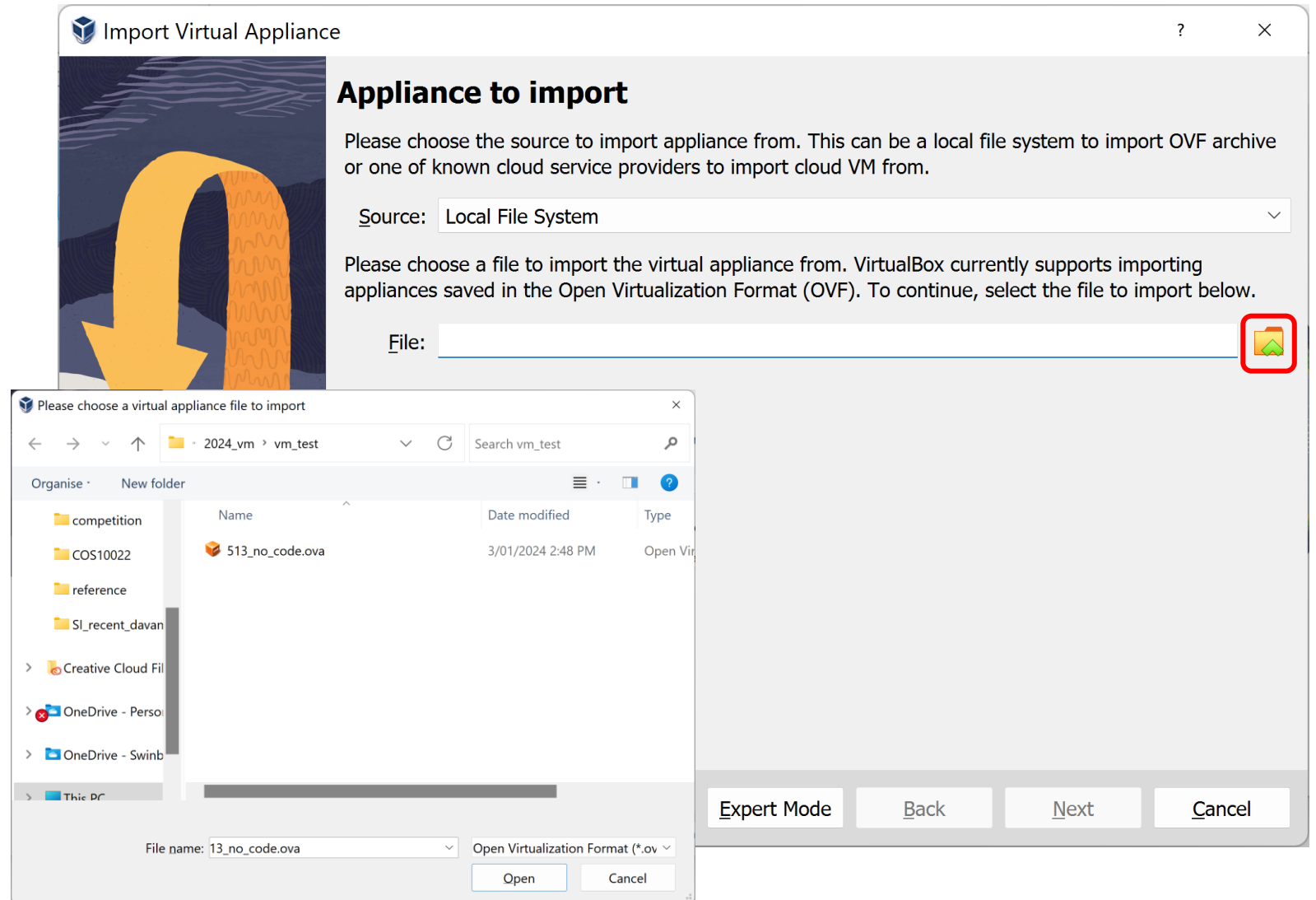
2. Download the VM Image in Canvas.

Module → COS20028 Tools → Cloudera VM

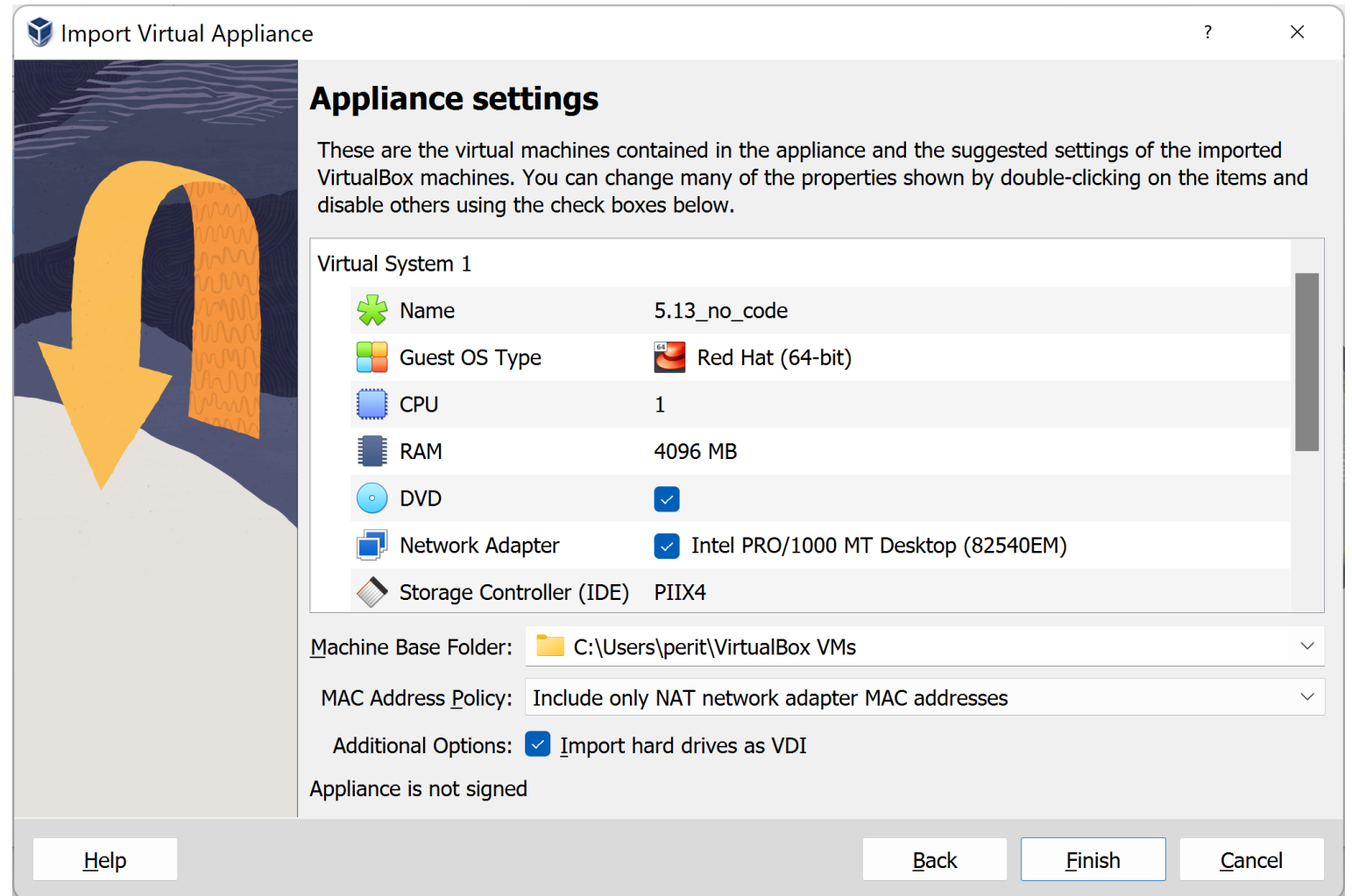
Import the VM Image



Import VM Image (2)

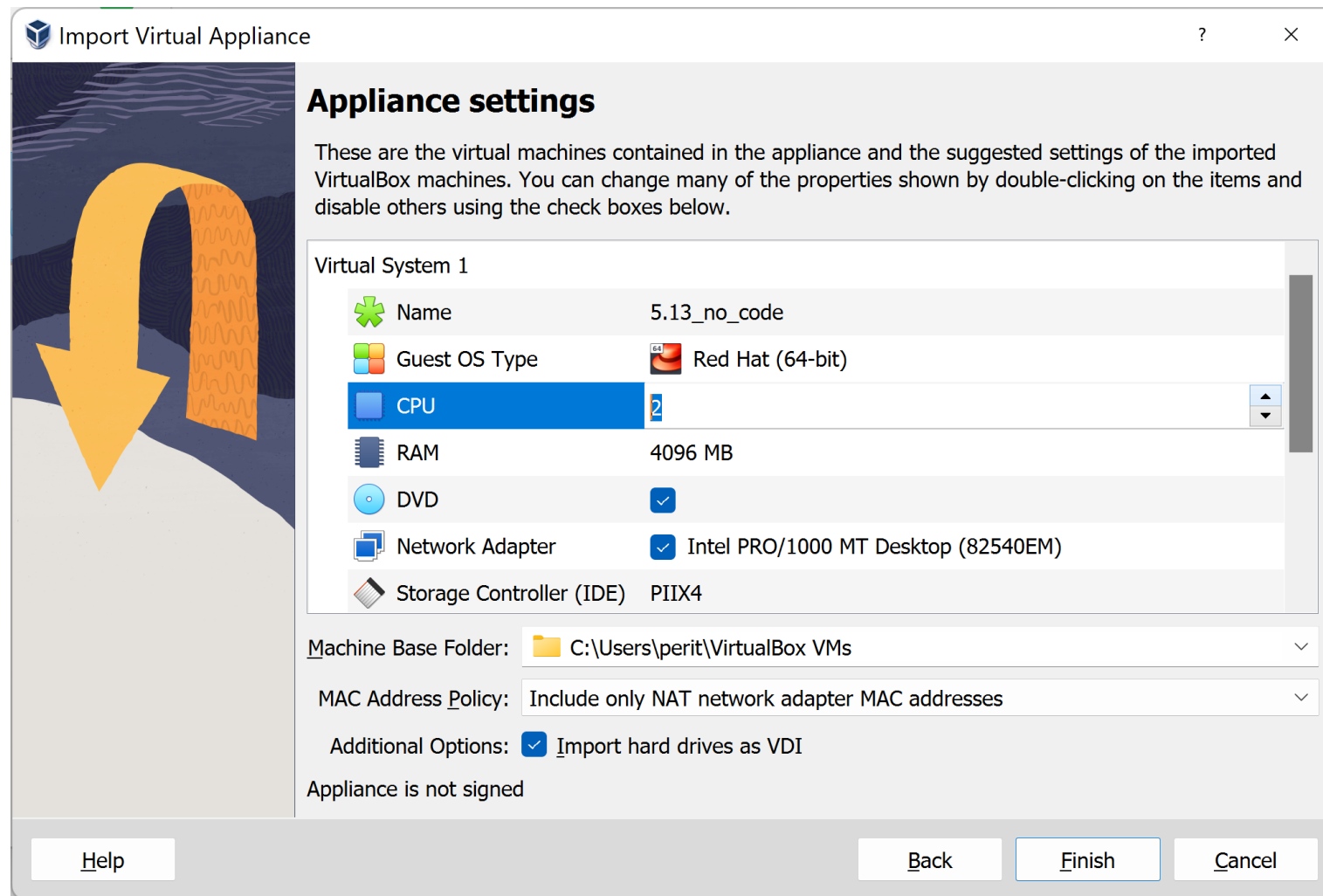


Review/Update VM Hardware Resource



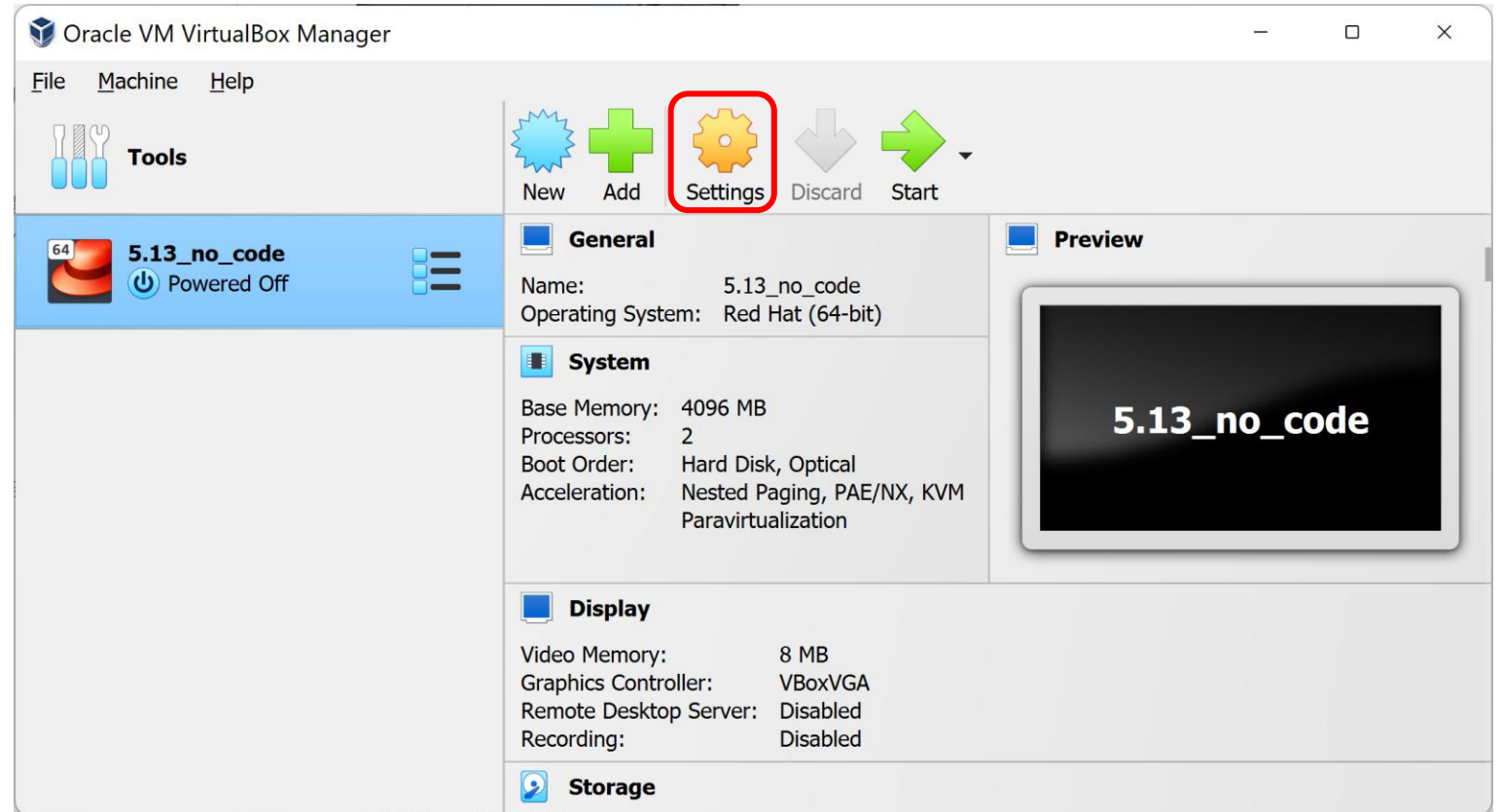
Review/Update VM Hardware Resource (2)

- You can optionally decide whether to increase the number of CPUs to 2 by double click on the corresponding row.
- If your physical machine is not very powerful, keep the virtual CPU count at 1.



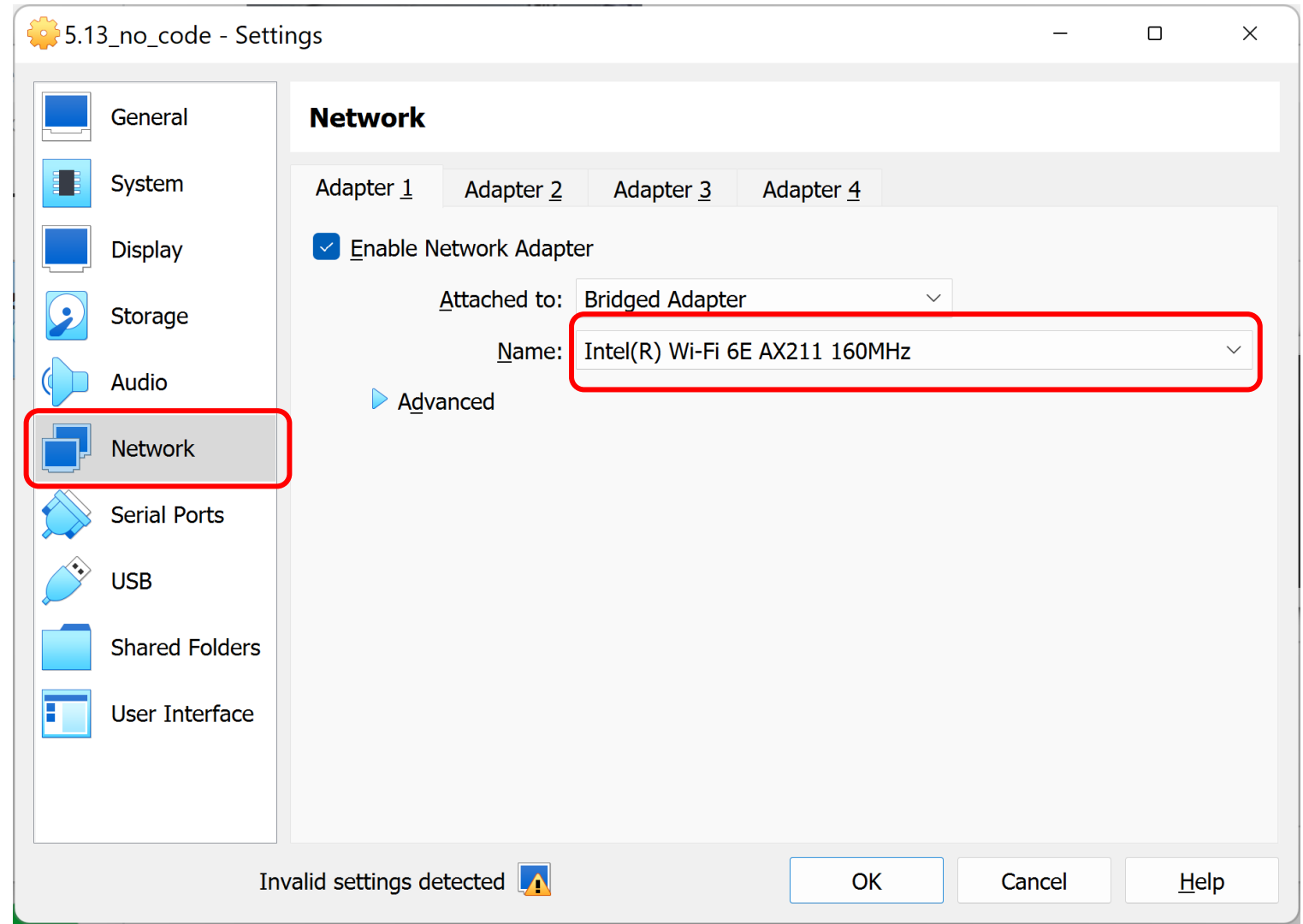
Update Network Setting

- The image was built on a machine which is different from yours.
- Click “Settings” to update the network configuration.



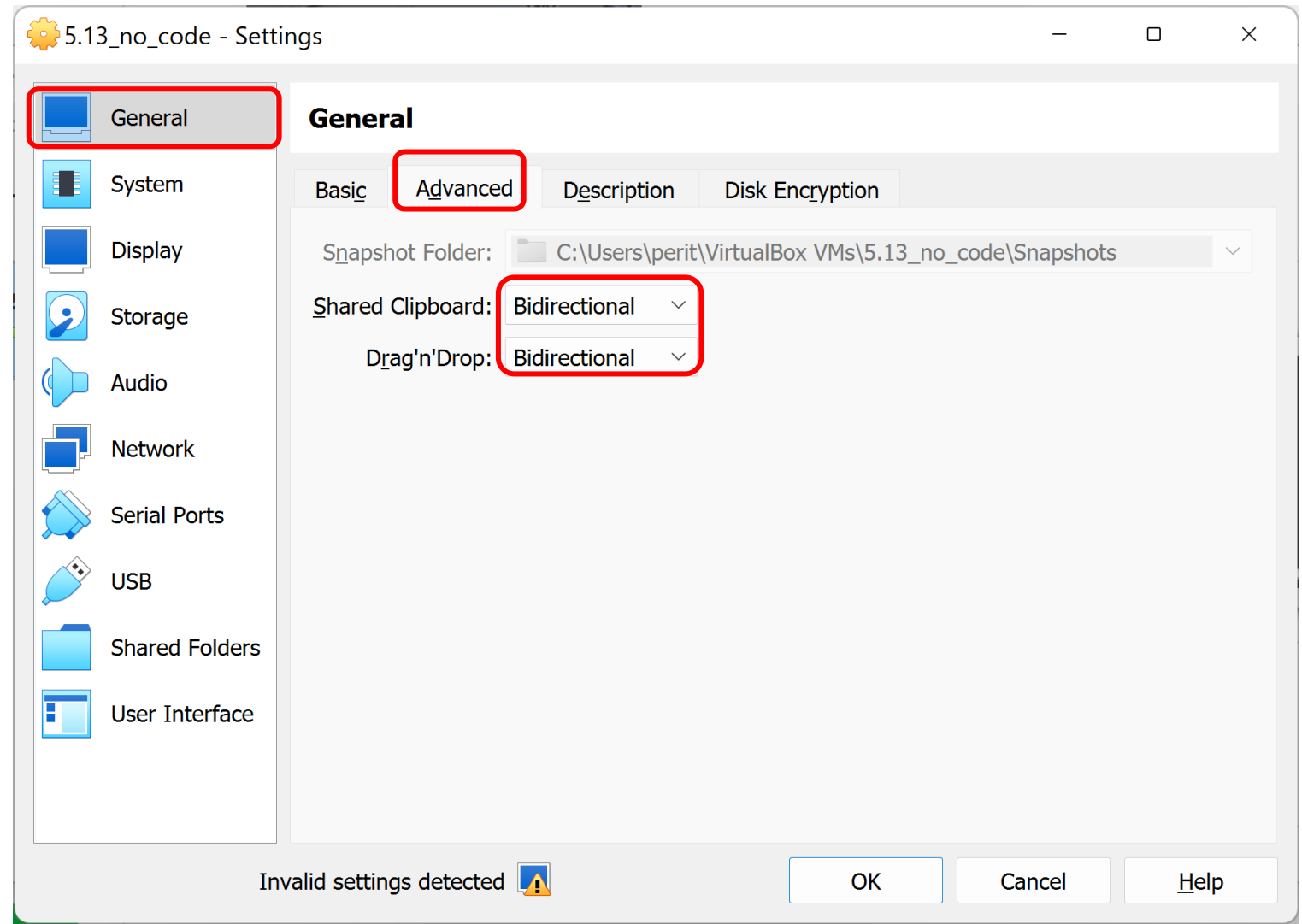
Update Network Setting

- The image was built on a machine which is different from yours.
- Click “Settings” to update the network configuration.
- Make sure the “Name” is selected to the correct network card.
- Click “OK” to save the change.

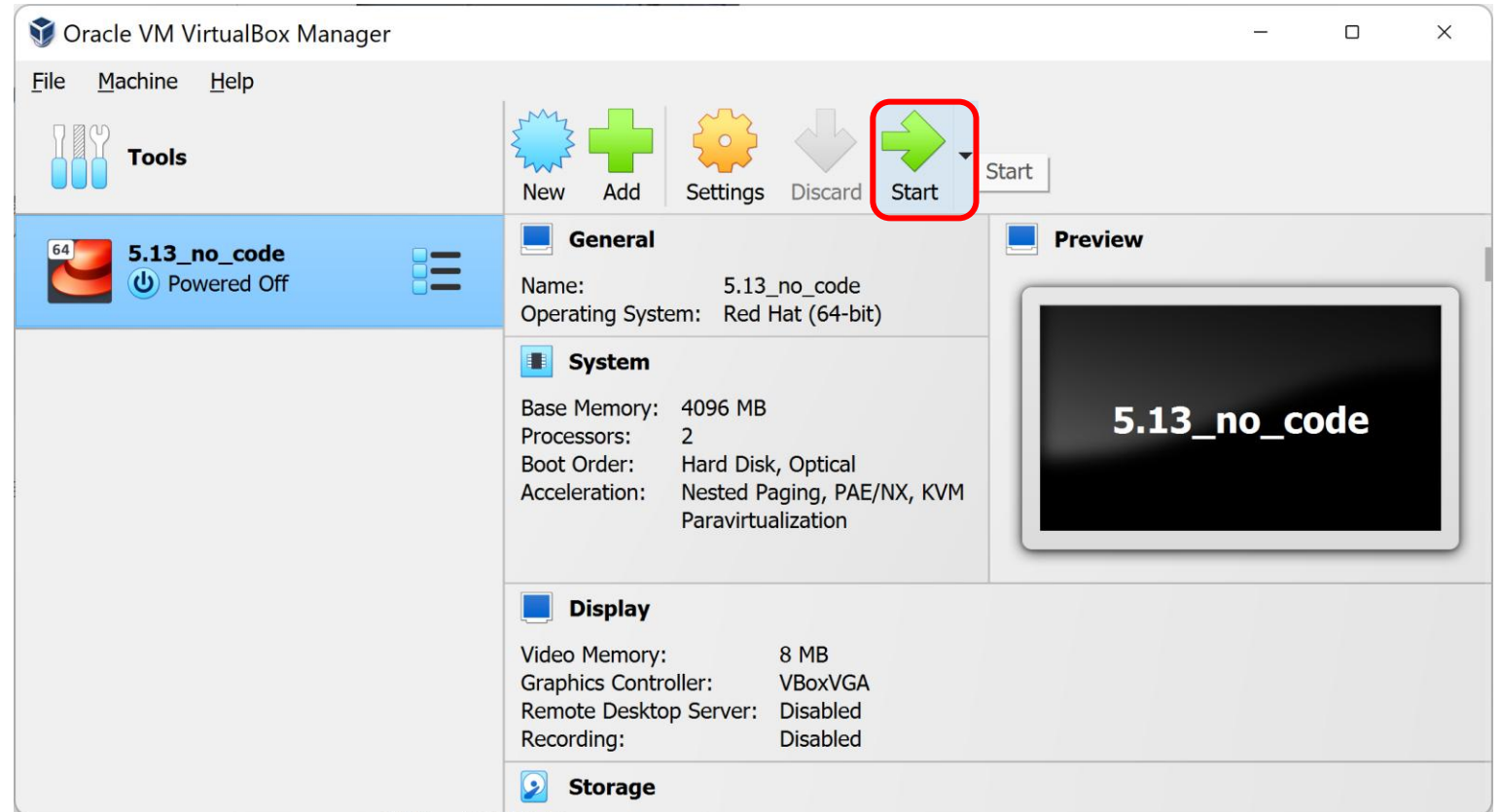


Update the Advanced Setting

- Enable the shared clipboard and drag'n'Drop helps you to work with the VM more conveniently.

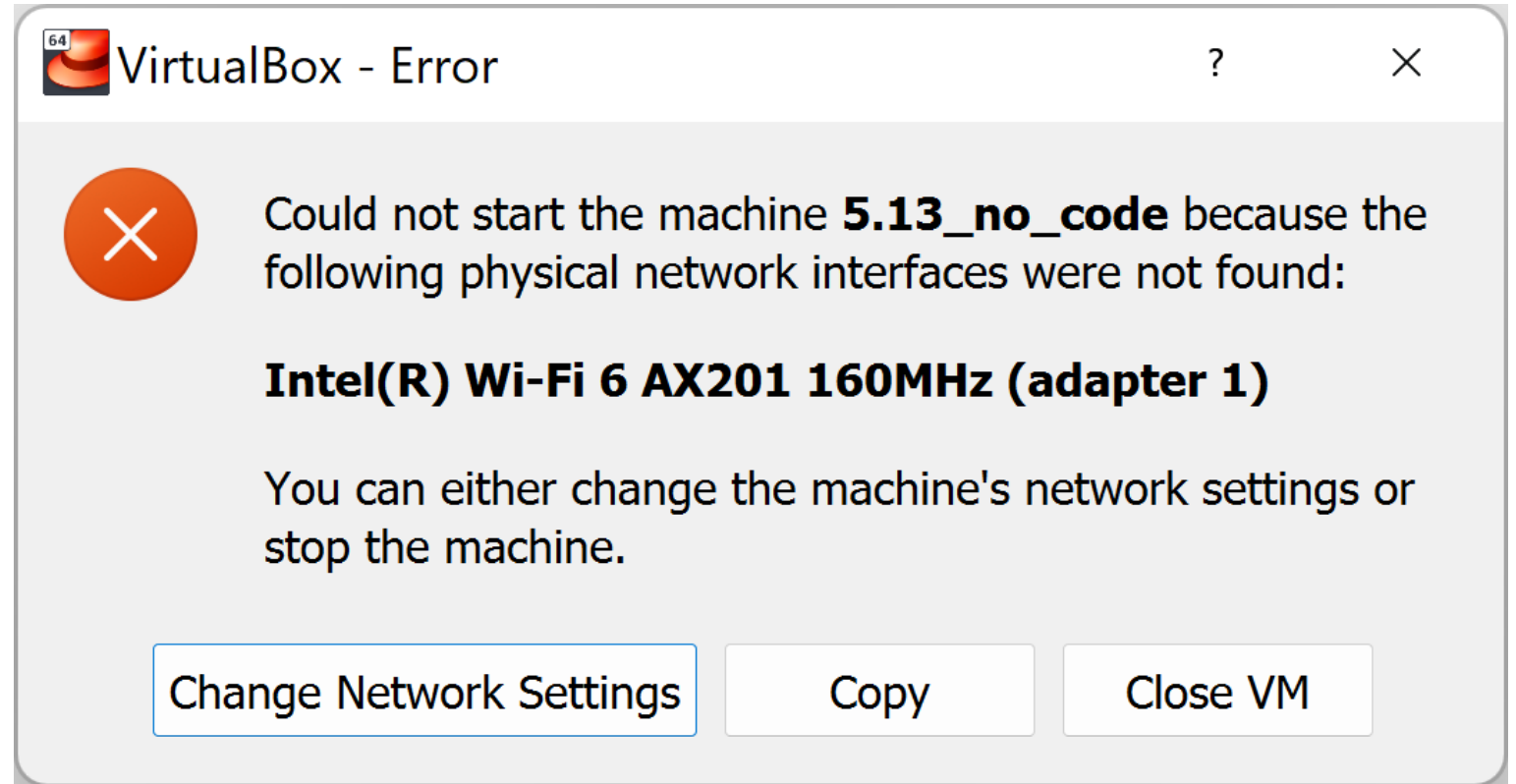


Start the Virtual Machine

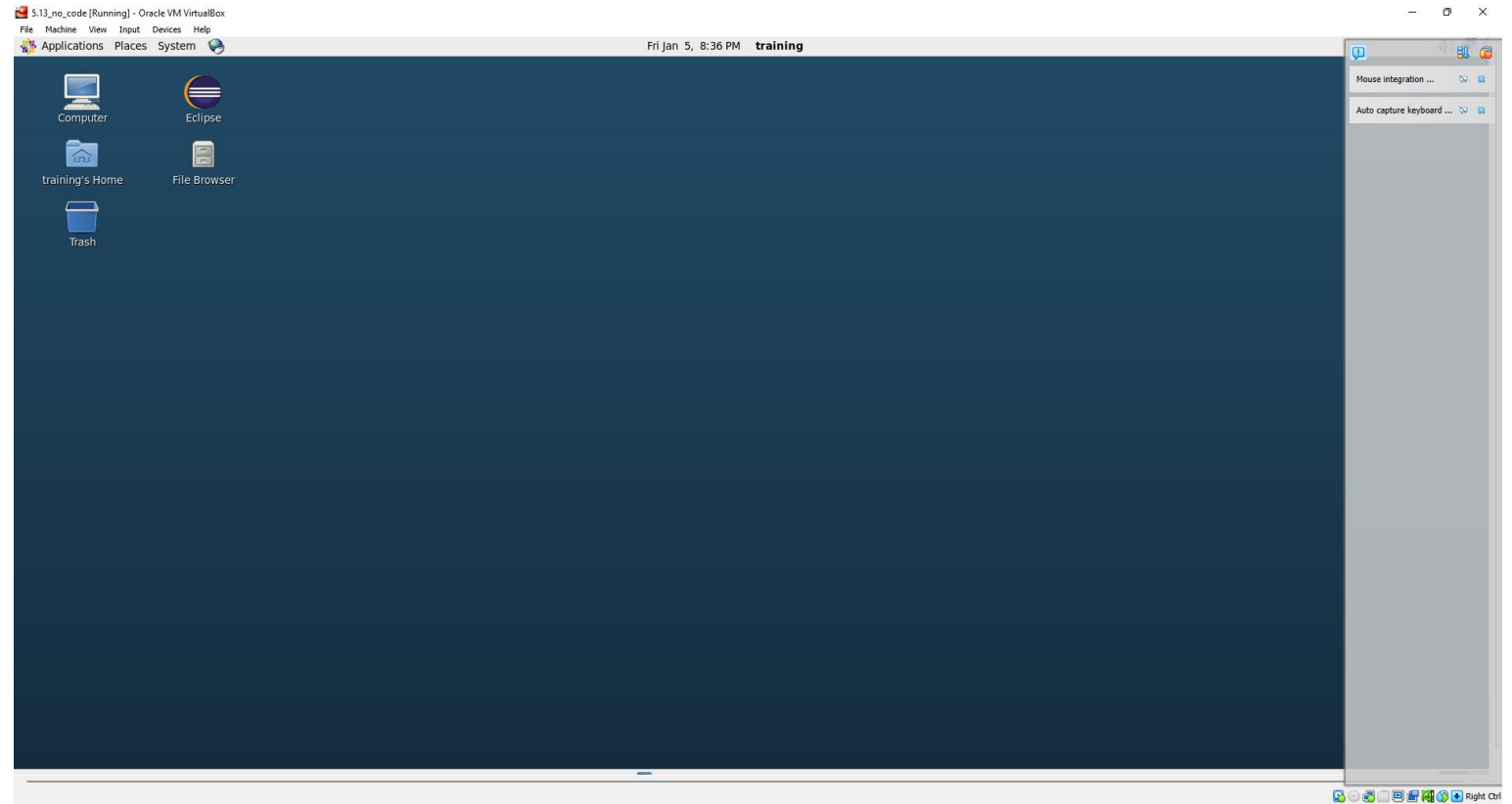


Encounter an Error?

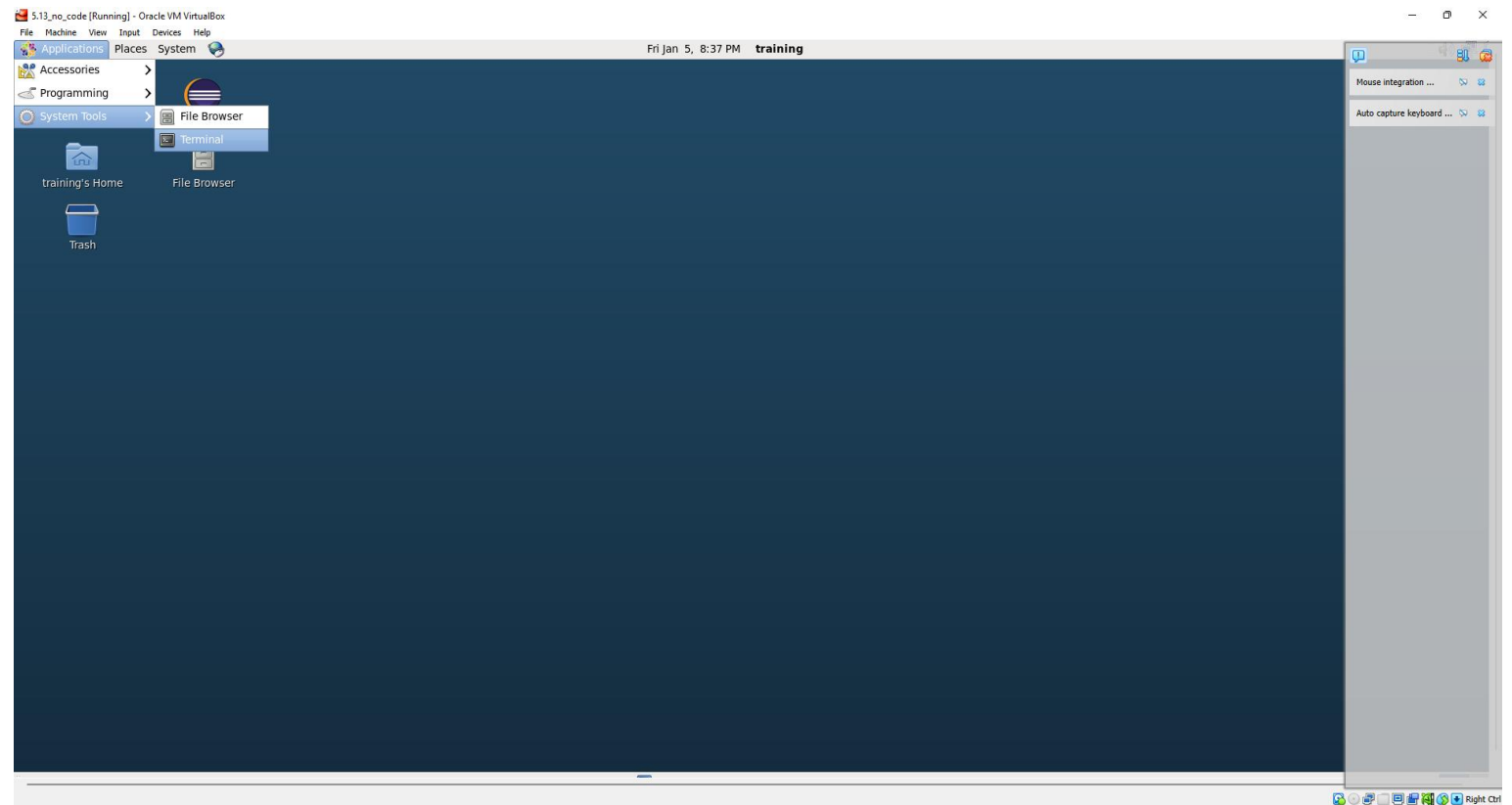
- It is normal if you encounter an error such as the physical network interface is not found.
- This image is built on a machine which is different from yours.
- The solution is to change the network settings.



Virtual Machine Environment



Open a Terminal



```
training@quickstart:~/training_materials/developer/data
File Edit View Search Terminal Help
[training@quickstart ~]$ cd ~/training_materials/developer/data/
[training@quickstart data]$ ls
2022_access_log.gz      movielens.readme      nameyeartestdata
access_log.gz           movielens-small.sql   shakespeare-stream.tar.gz
bible.tar.gz            movielens.sql         shakespeare.tar.gz
invertedIndexInput.tgz  movie_metadata.csv
[training@quickstart data]$ hdfs dfs -ls
[training@quickstart data]$ hdfs dfs -put shakespeare.tar.gz
```

Upload File to HDFS

- **Switch to the source folder and list the content:**
 - `cd ~/training_materials/developer/data/`
 - `ls`
- **Check what is in your Hadoop folder:**
 - `hdfs dfs -ls`
 - (By default, it checks the user's home directory.)
(In this case: `/user/training`)
 - Displays nothing when the folder is empty.
- **Upload the file/folder:**
 - `hdfs dfs -put Shakespeare.tar.gz`

```
training@quickstart:~/training_materials/developer/data
File Edit View Search Terminal Help
[training@quickstart ~]$ cd ~/training_materials/developer/data/
[training@quickstart data]$ ls
2022_access_log.gz      movielens.readme      nameyeartestdata
access_log.gz           movielens-small.sql   shakespeare-stream.tar.gz
bible.tar.gz            movielens.sql         shakespeare.tar.gz
invertedIndexInput.tgz  movie_metadata.csv
[training@quickstart data]$ hdfs dfs -ls
[training@quickstart data]$ hdfs dfs -put shakespeare.tar.gz
[training@quickstart data]$ hdfs dfs -ls
Found 1 items
-rw-r--r--  1 training training    2080857 2024-01-05 21:09 shakespeare.tar.gz
[training@quickstart data]$
```

Check Hadoop Content

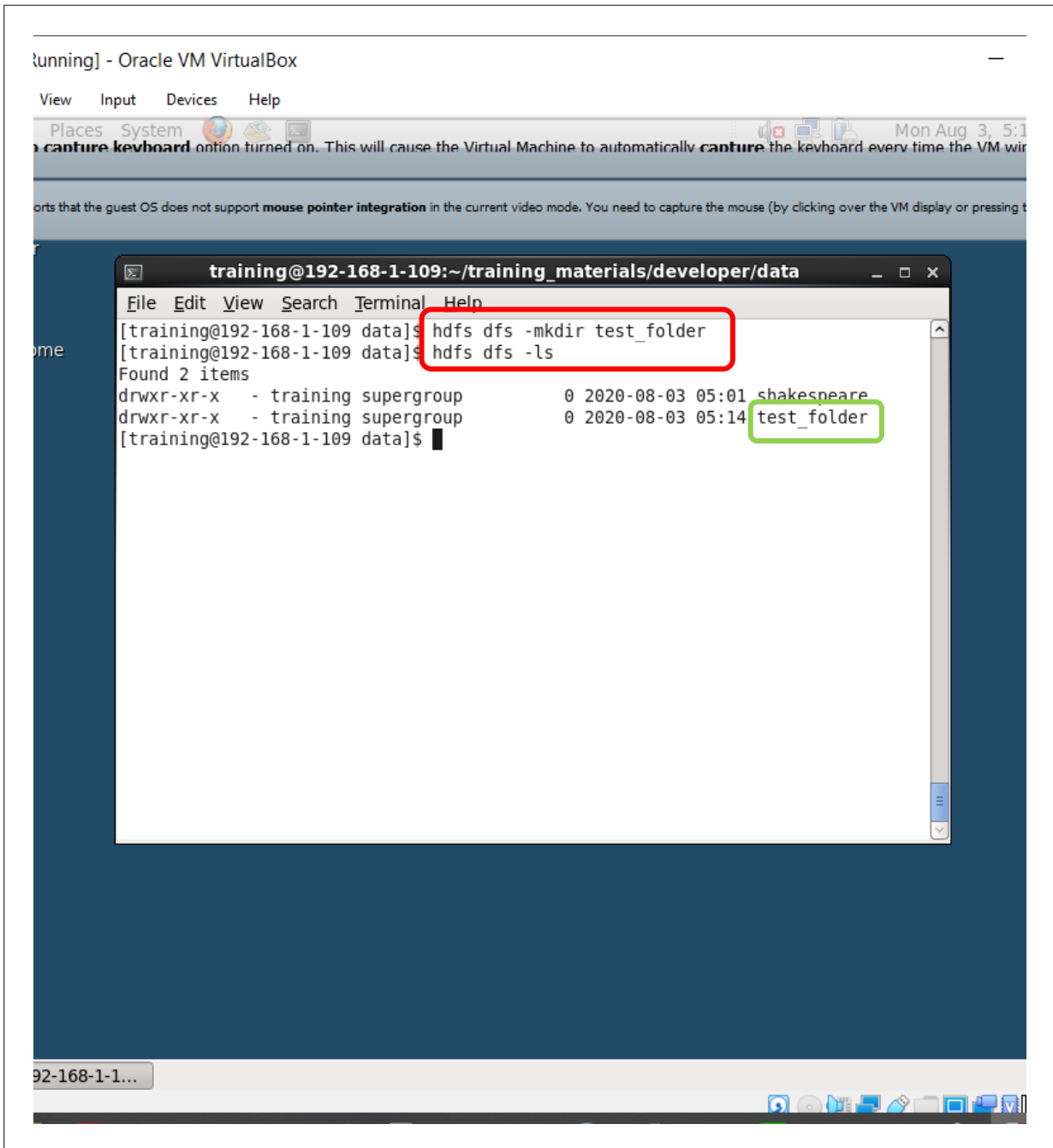
- Check what is in your Hadoop folder:
 - `hdfs dfs -ls`
 - You'll find the uploaded compressed file is there.

```
training@quickstart:~/training_materials/developer/data
File Edit View Search Terminal Help
[training@quickstart data]$ hdfs fsck /user/training
Connecting to namenode via http://quickstart.cloudera:50070/fsck?ugi=training&path=%2Fuser%2Ftraining
FSCK started by training (auth:SIMPLE) from /127.0.0.1 for path /user/training at Fri Jan 05 21:14:22 PST 2024
.Status: HEALTHY
Total size:      2080857 B
Total dirs:      1
Total files:      1
Total symlinks:    0
Total blocks (validated): 1 (avg. block size 2080857 B)
Minimally replicated blocks: 1 (100.0 %)
Over-replicated blocks: 0 (0.0 %)
Under-replicated blocks: 0 (0.0 %)
Mis-replicated blocks: 0 (0.0 %)
Default replication factor: 1
Average block replication: 1.0
Corrupt blocks: 0
Missing replicas: 0 (0.0 %)
Number of data-nodes: 1
Number of racks: 1
FSCK ended at Fri Jan 05 21:14:22 PST 2024 in 10 milliseconds

The filesystem under path '/user/training' is HEALTHY
[training@quickstart data]$
```

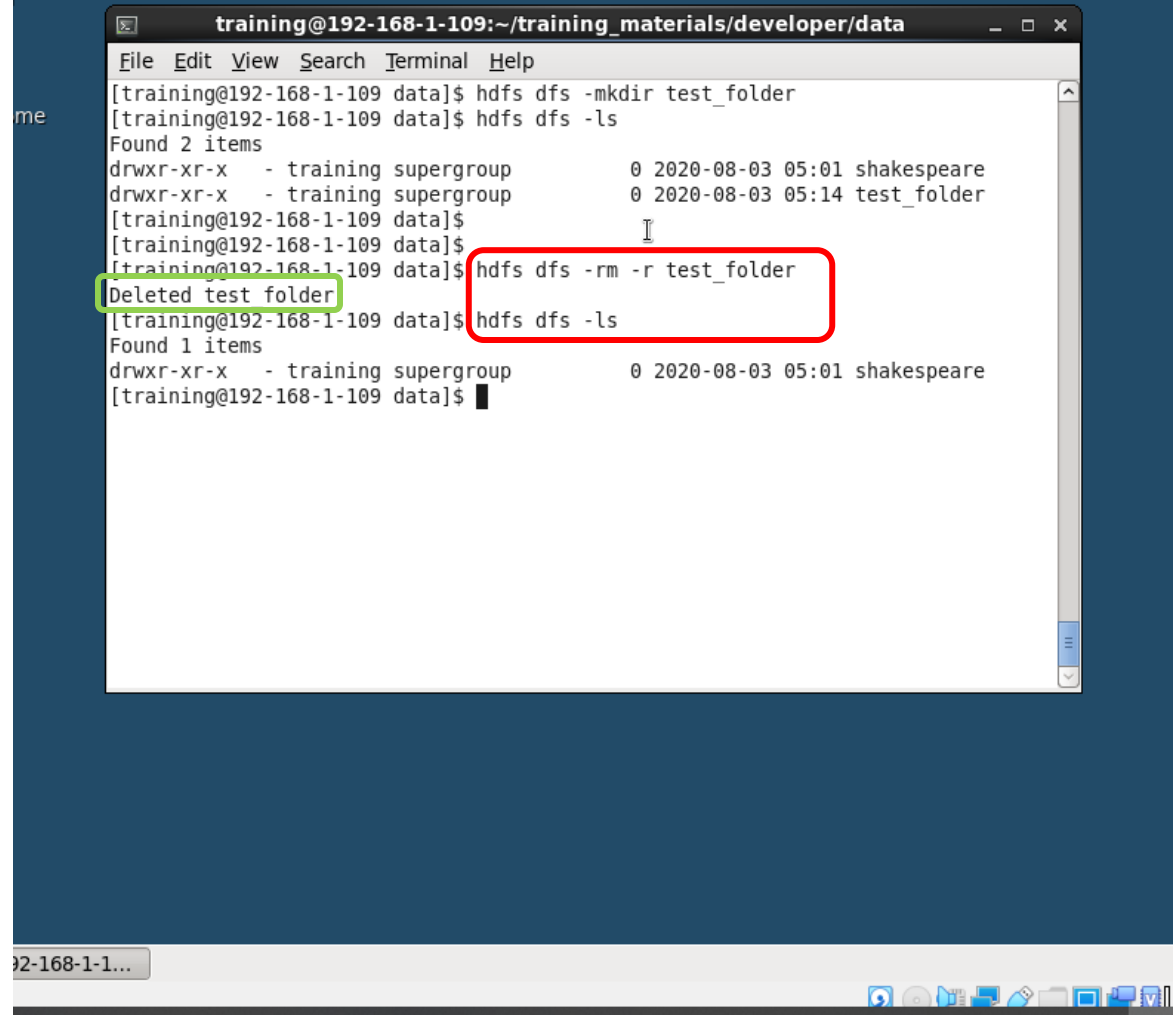
Check HDFS Healthy Status

- **hdfs fsck /user/training**



Create Folder

- `hdfs dfs -mkdir test_folder`
- `hdfs dfs -ls`



```
training@192-168-1-109:~/training_materials/developer/data
File Edit View Search Terminal Help
[training@192-168-1-109 data]$ hdfs dfs -mkdir test_folder
[training@192-168-1-109 data]$ hdfs dfs -ls
Found 2 items
drwxr-xr-x - training supergroup      0 2020-08-03 05:01 shakespeare
drwxr-xr-x - training supergroup      0 2020-08-03 05:14 test_folder
[training@192-168-1-109 data]$
[training@192-168-1-109 data]$ hdfs dfs -rm -r test_folder
Deleted test folder
[training@192-168-1-109 data]$ hdfs dfs -ls
Found 1 items
drwxr-xr-x - training supergroup      0 2020-08-03 05:01 shakespeare
[training@192-168-1-109 data]$
```

Remove Folder

- `hdfs dfs -rm -r test_folder`
- `hdfs dfs -ls`

[Running] - Oracle VM VirtualBox

```
View Input Devices Help
s Places System Mon Aug 3, 5
root@192-168-1-109:~
w Search Terminal Help
9:07 INFO DataNode.c
ytes: 1798560, op: HD
: 0, srvID: DS-460392
7.0.0.1-1386722652683
9:07 INFO DataNode.c
ytes: 59440, op: HDFS
0, srvID: DS-46039223
0.0.1-1386722652683:b
9:07 INFO DataNode.c
ytes: 1490591, op: HD
: 0, srvID: DS-460392
7.0.0.1-1386722652683
9:07 INFO DataNode.c
ytes: 270236, op: HDFS
0, srvID: DS-4603922
.0.0.1-1386722652683:
9:07 INFO DataNode.c
ytes: 1766132, op: HD
: 0, srvID: DS-460392
7.0.0.1-1386722652683

[training@192-168-1-109:~]
File Edit View Search Terminal Help
[training@192-168-1-109 ~]$ cd ~
[training@192-168-1-109 ~]$ hdfs dfs -get shakespeare cp_shakespeare
[training@192-168-1-109 ~]$ ls
cp_shakespeare      lib      templates
Desktop             Music    training_materials
Documents            Pictures Videos
Downloads            Public   workspace
eclipse             scripts
kiji-bento-albacore-1.0.5-release.tar.gz src
[training@192-168-1-109 ~]$
```

Download Files from Hadoop

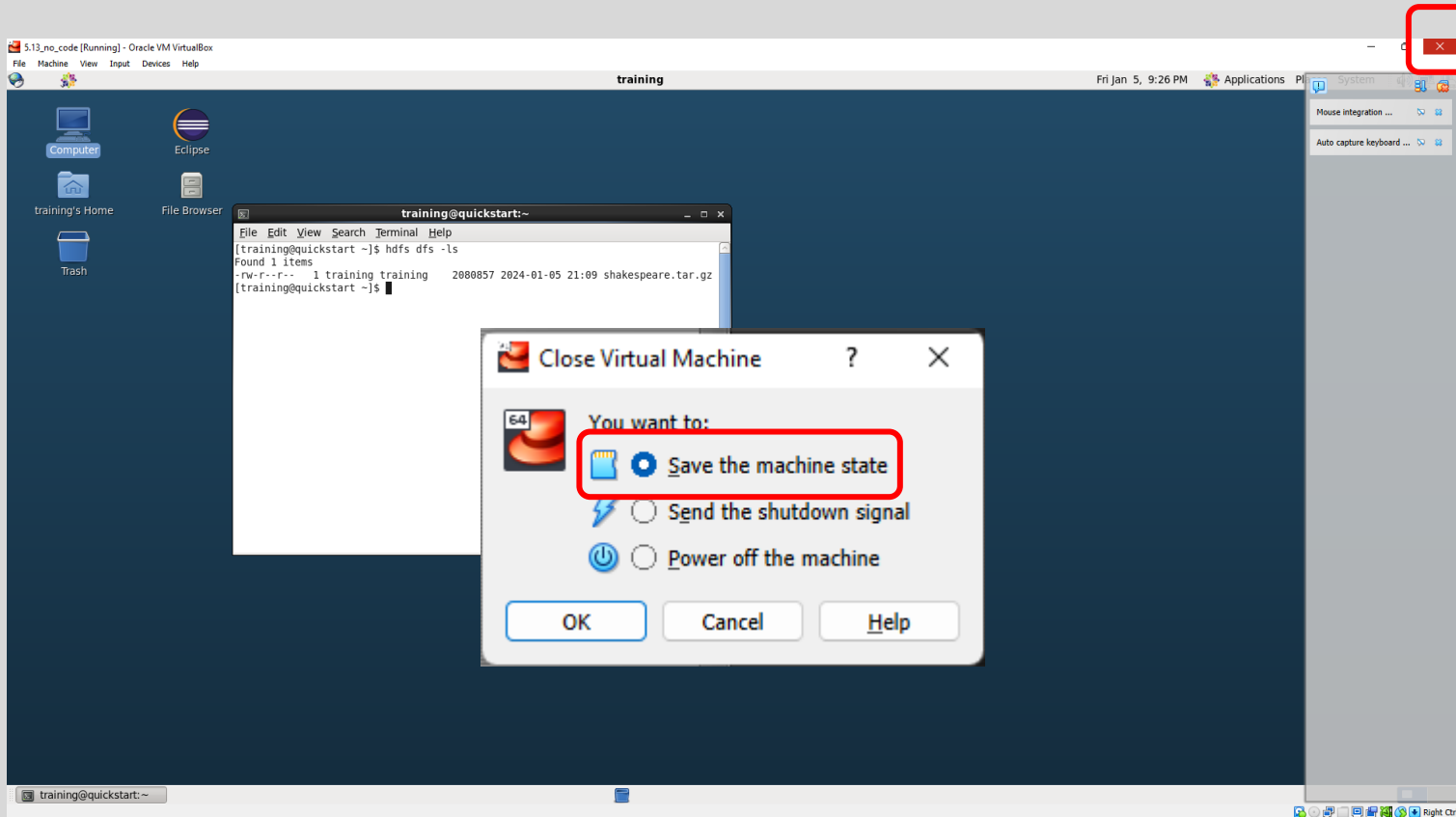
- `cd ~`
- `hdfs dfs -get shakespeare cp_shakespeare`
- `ls`

Password for Entering the System

- If you got locked out by the screen saver, use the password “**training**” to get back in again.



Freeze the Machine State



- Freeze the state of the machine instead of powering it down.
- This will help you get everything loaded much faster the next time you enter the virtual machine.